ORF 524 — Midterm Exam

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Contents

Notes:

- Duration of examination: 90 minutes.
  An additional 15 minutes are given to upload your solutions to Gradescope.
- All parts are equally weighted (5 points).
  Please provide as much detail as possible in your answers.
  Answers without proper justification will not receive full credit.
- Please start each question on a separate page.
  A single PDF file must be submitted to Gradescope.
- Only one page of personal notes are allowed during the examination.
  This is otherwise a closed-book and individual examination.

1 Question 1: Sufficiency and Completeness (25 points)

Let $X \sim \mathbb{P}_\theta$ where $\theta \in \Theta$. Let $S(X)$ be a statistic and m be a measurable function.

1. Show that if $m(S(X))$ is a sufficient statistic then $S(X)$ is a sufficient statistic.
2. Show that if $S(X)$ is sufficient and the inverse of m exists and is measurable then $m(S(X))$ is sufficient.
3. Give an example of where both $S(X)$ and $m(S(X))$ are sufficient but m has no inverse.
4. Show that if $S(X)$ is complete then $m(S(X))$ is complete.
5. Supposing a minimal sufficient statistic exists, show that if $S(X)$ is complete sufficient and bounded then $S(X)$ is minimal sufficient.

2 Question 2: Finite-Sample Statistical Inference (40 points)

Let $X = (X_i : 1 \leq i \leq n)$ where $X_i \sim \mathbb{P}_\theta$ i.i.d. with Lebesgue density

$$f(x;\theta) = \frac{2x}{\theta^4} \mathbb{1}\{0 \leq x \leq \theta^2\}$$

for $\theta > 0$, where $\mathbb{1}$ is the indicator function.

1. Find $F(x;\theta) = \mathbb{P}_\theta(X_i \leq x)$, the distribution function of X_i.

2. Consider the following estimators:

$$\hat{\theta} = \frac{1}{n}\sum_{i=1}^{n} \sqrt{X_i} \qquad \text{and} \qquad \hat{\theta}_{\texttt{ML}} = \max_{1 \leq i \leq n} \sqrt{X_i}.$$

   (a) Propose an unbiased estimator of θ based on $\hat{\theta}$ and denote it by $\hat{\theta}_{\texttt{U}}$. Can $\hat{\theta}_{\texttt{U}}$ be interpreted as a method of moments estimator?

   (b) Show that $\hat{\theta}_{\texttt{ML}}$ is the maximum likelihood estimator for θ and that it is a complete sufficient statistic.

   (c) Find a uniformly minimum variance unbiased (UMVU) estimator $\hat{\theta}_{\texttt{UMVU}}$ for θ.

   (d) Calculate $\mathbb{V}_\theta\left[\hat{\theta}_{\texttt{U}}\right]$ and $\mathbb{V}_\theta\left[\hat{\theta}_{\texttt{UMVU}}\right]$ and give a necessary and sufficient condition under which $\mathbb{V}_\theta[\hat{\theta}_{\texttt{U}}] > \mathbb{V}_\theta[\hat{\theta}_{\texttt{UMVU}}]$.

3. Consider the following hypothesis testing problem:

$$\mathsf{H}_0 : \theta \leq \theta_0 \qquad \text{vs.} \qquad \mathsf{H}_1 : \theta > \theta_0$$

   where $\theta_0 > 0$. We use a test that rejects H_0 if and only if $\hat{\theta}_{\texttt{ML}}/\theta_0 > c$ for some critical value c.

   (a) Find c so that the test has 5% size.

   (b) Derive and sketch a plot of the power function $\beta(\theta)$.

   (c) Propose a 95% one-sided confidence interval for θ based on this test.